Name: Pradeepa. S Register Number: 250701531

In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

In [3]:
```
data=pd.read_csv('Iris.csv')
data
```

Out[3]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

In [4]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    str    
dtypes: float64(4), str(1)
memory usage: 7.2 KB
```

In [5]:
```
data.describe()
```

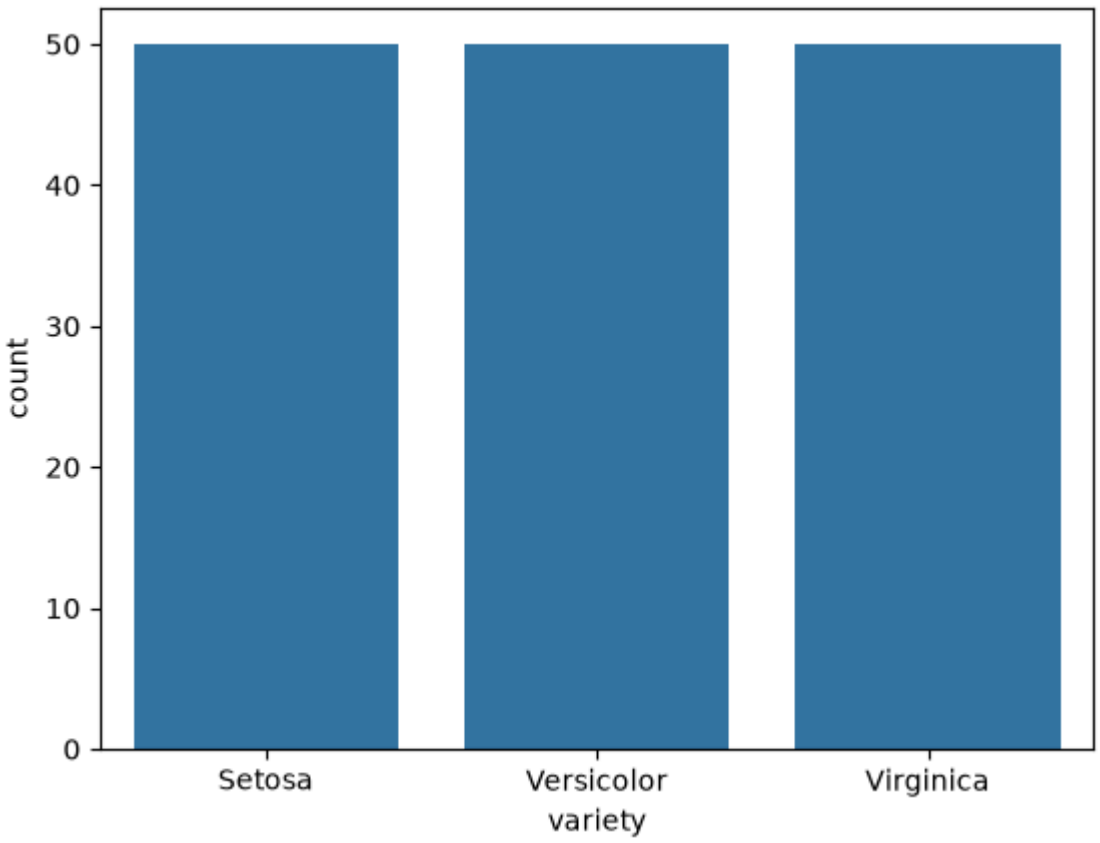
Out[5]:

	sepal.length	sepal.width	petal.length	petal.width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [6]:
```
data.value_counts('variety')
```

Out[6]:
```
variety
Setosa        50
Versicolor    50
Virginica     50
Name: count, dtype: int64
```

In [7]:
```
sns.countplot(x='variety',data=data,)
plt.show()
```



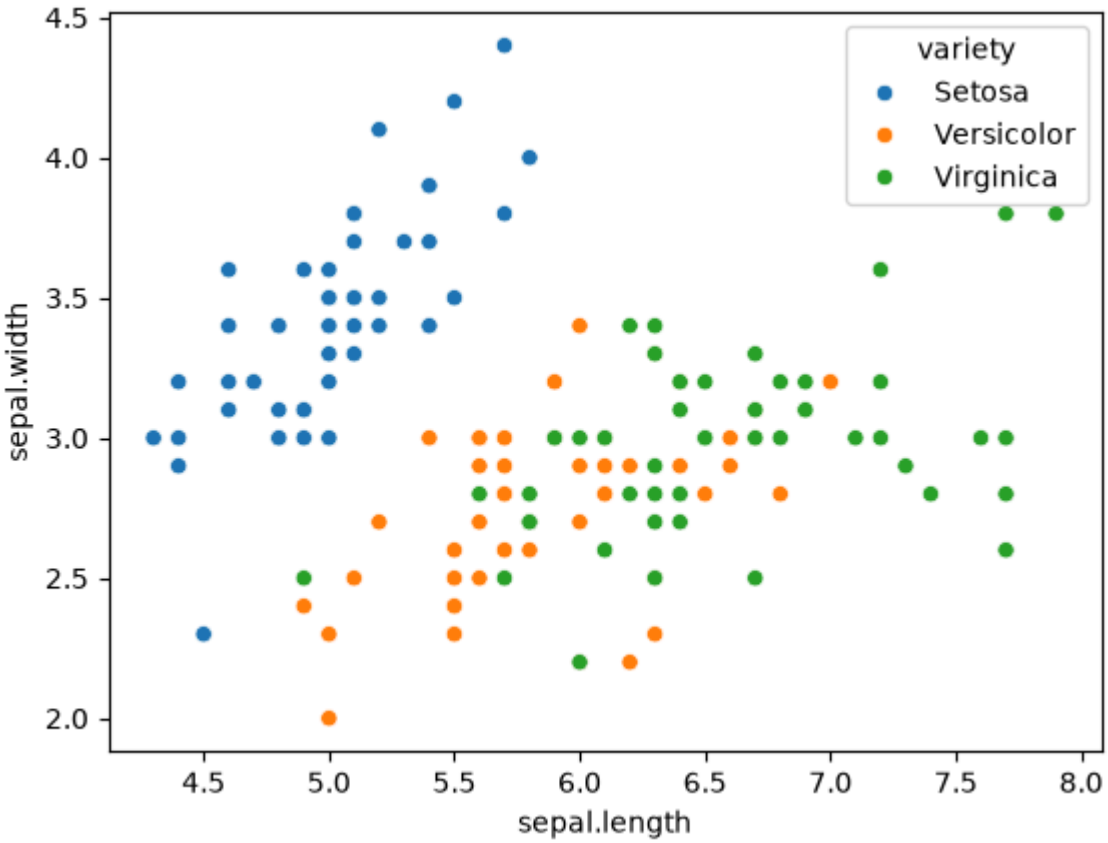
In [8]:
```
dummies=pd.get_dummies(data.variety)
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axi
FinalDataset.head()
```

Out[8]:

	Setosa	Versicolor	Virginica	sepal.length	sepal.width	petal.length	petal.width
0	True	False	False	5.1	3.5	1.4	0.2
1	True	False	False	4.9	3.0	1.4	0.2
2	True	False	False	4.7	3.2	1.3	0.2
3	True	False	False	4.6	3.1	1.5	0.2
4	True	False	False	5.0	3.6	1.4	0.2

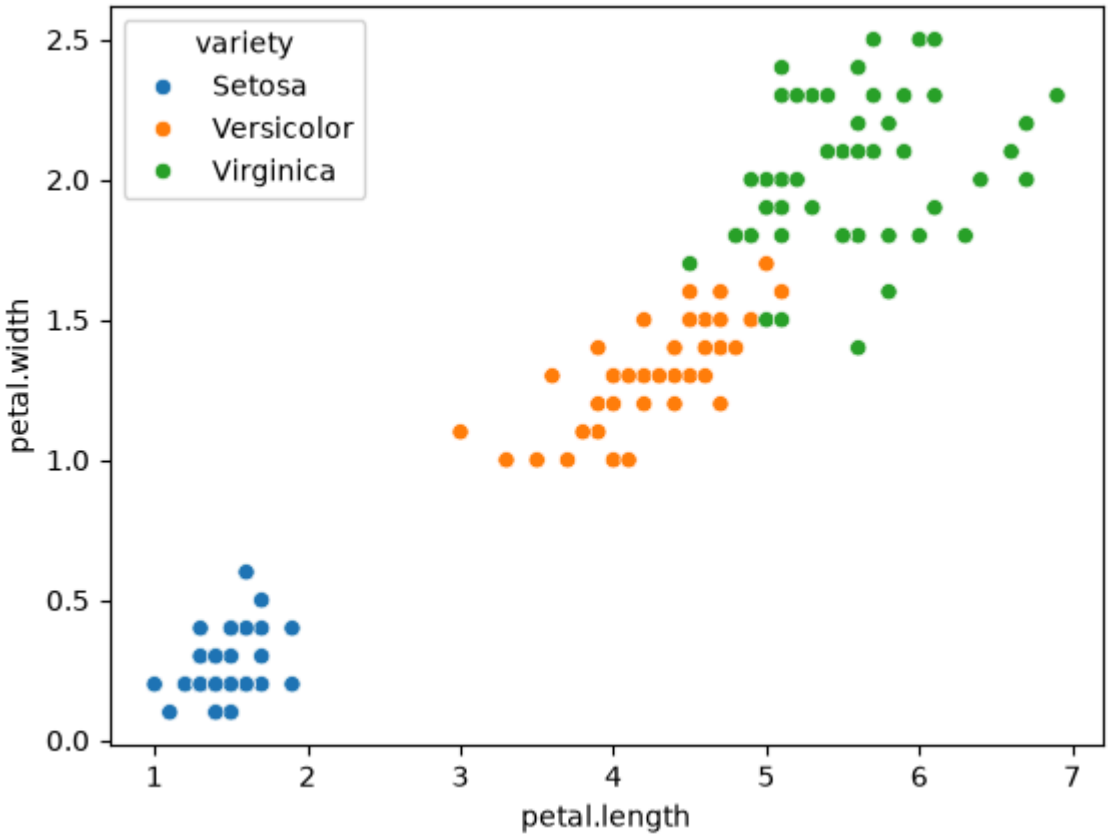
In [11]:
```
sns.scatterplot(x='sepal.length',y='sepal.width',hue='variety',data=data)
```

Out[11]:
```
<Axes: xlabel='sepal.length', ylabel='sepal.width'>
```

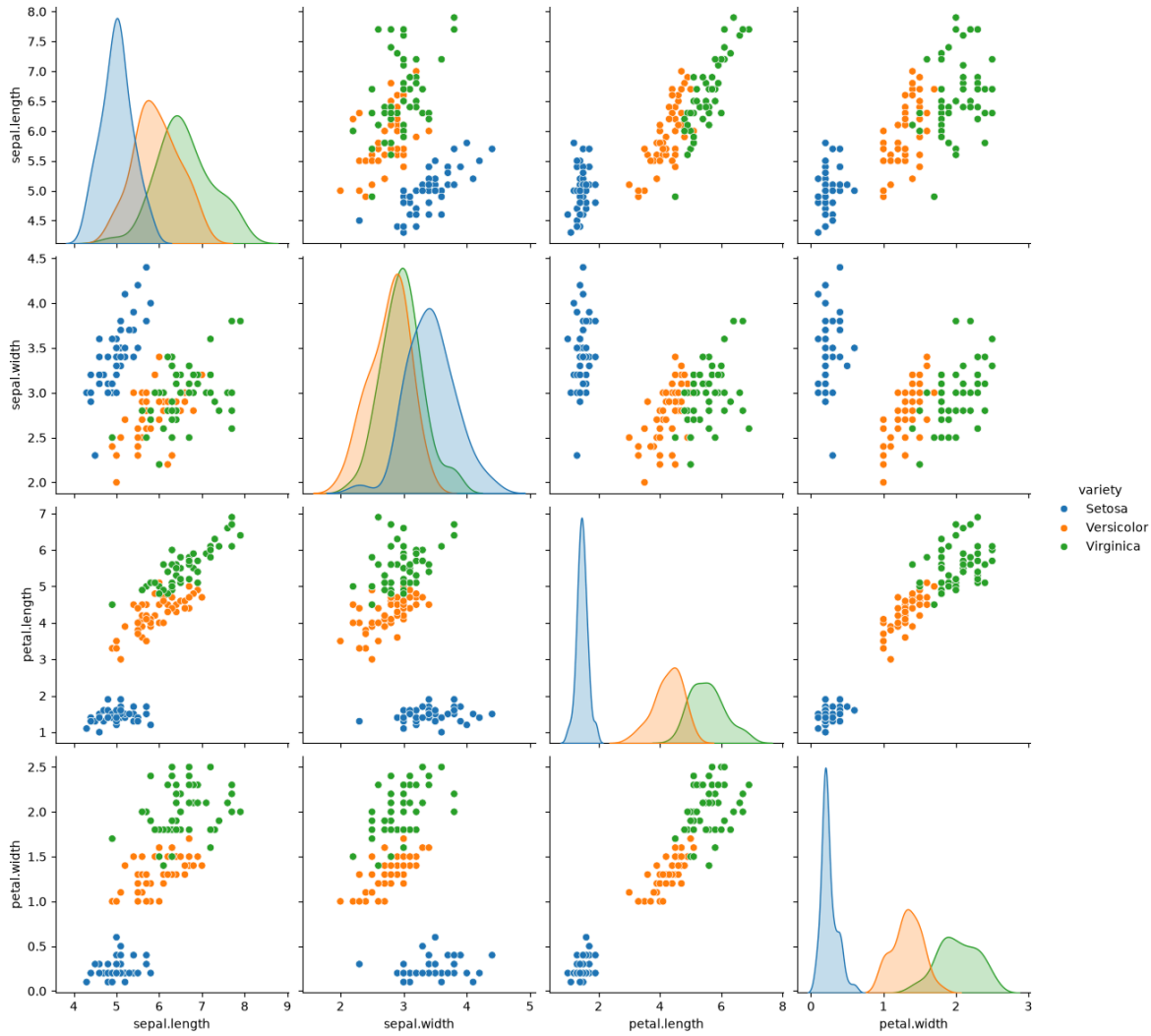


In [13]:
```
sns.scatterplot(x='petal.length',y='petal.width',hue='variety',data=data)
```

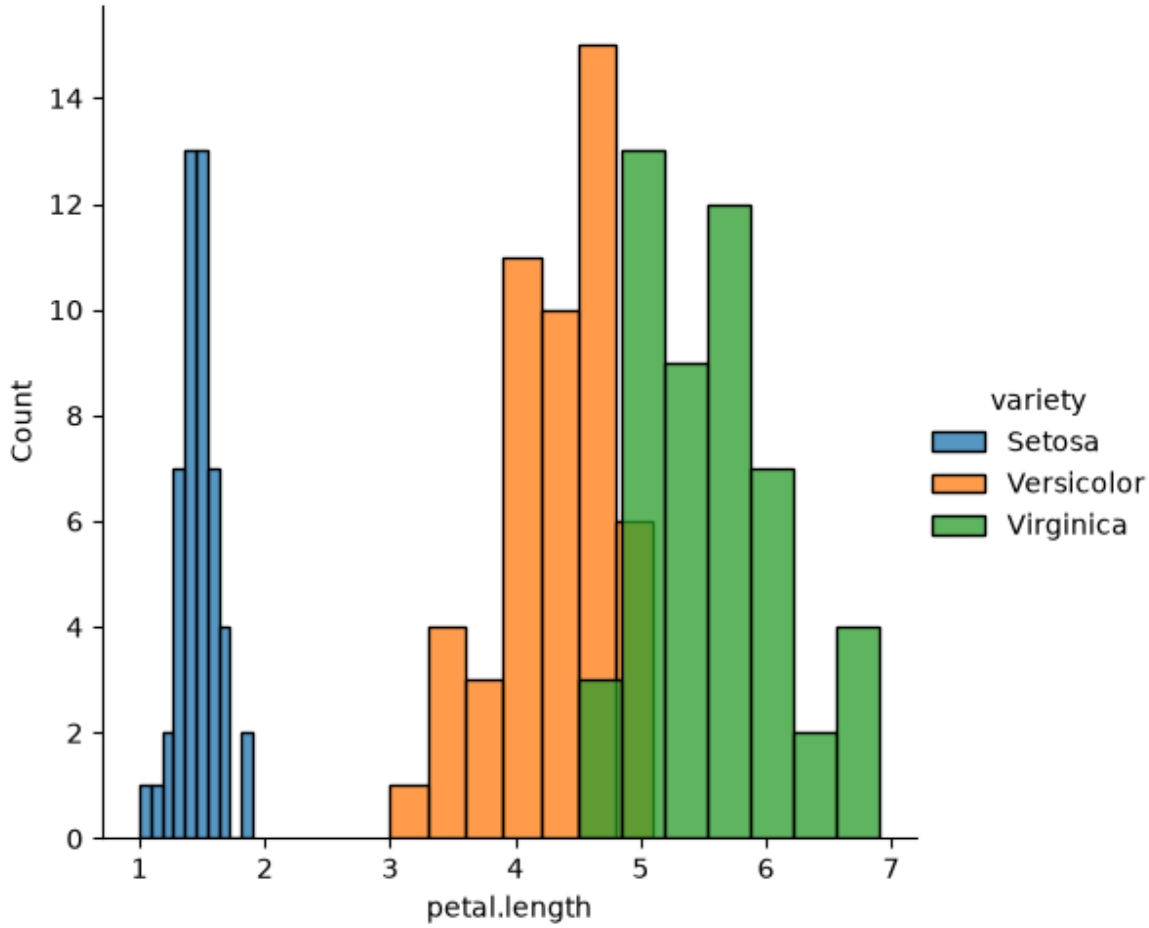
Out[13]:
```
<Axes: xlabel='petal.length', ylabel='petal.width'>
```



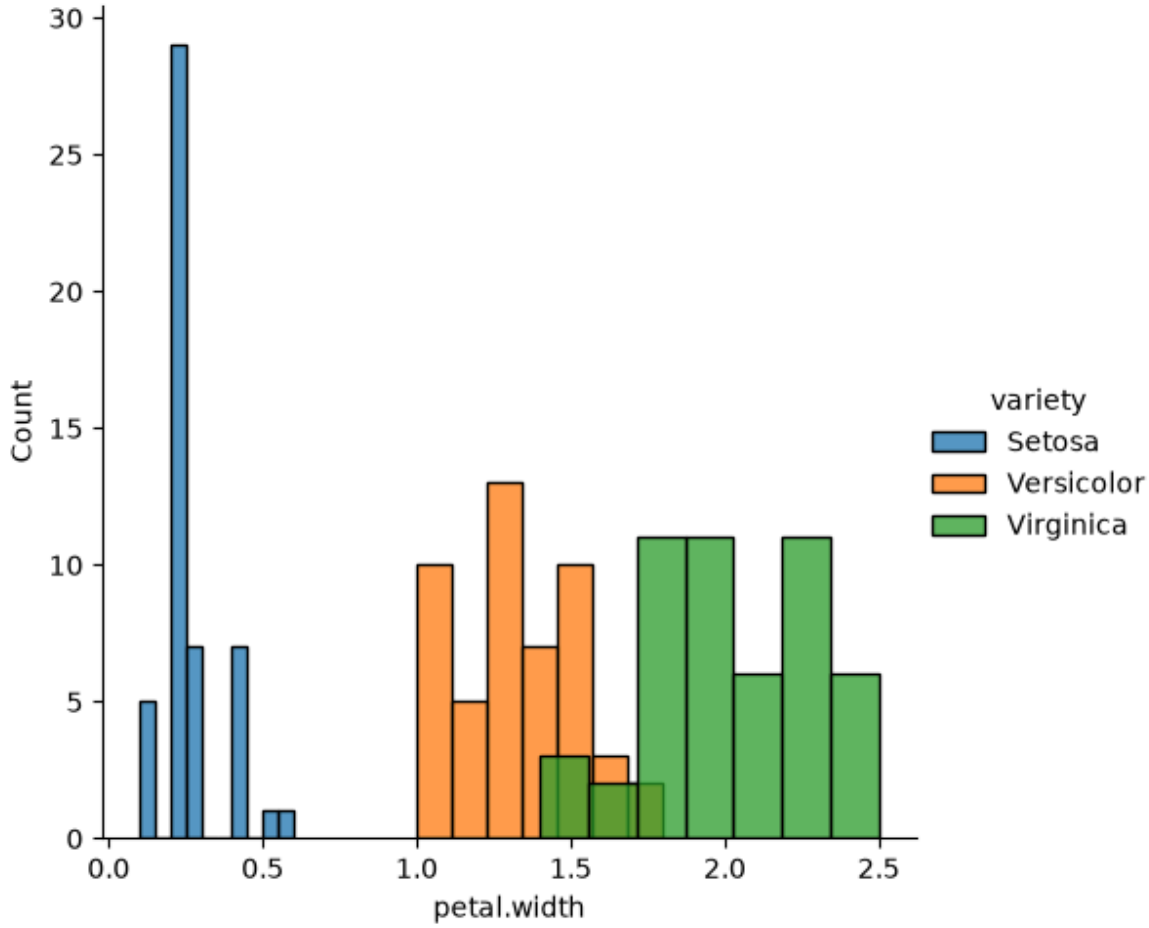
In [14]:
```
sns.pairplot(data,hue='variety',height=3);
```



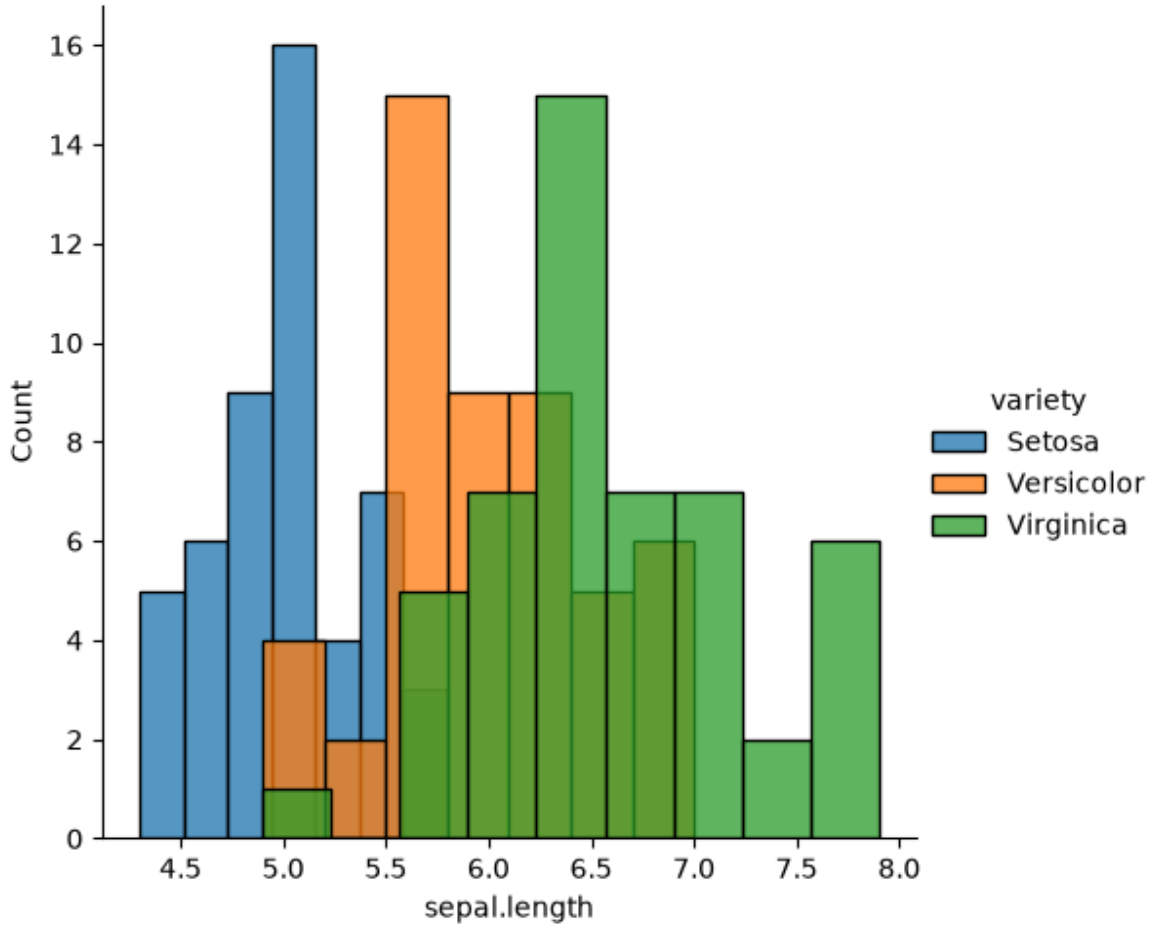
In [17]:
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'petal.length').add_
```



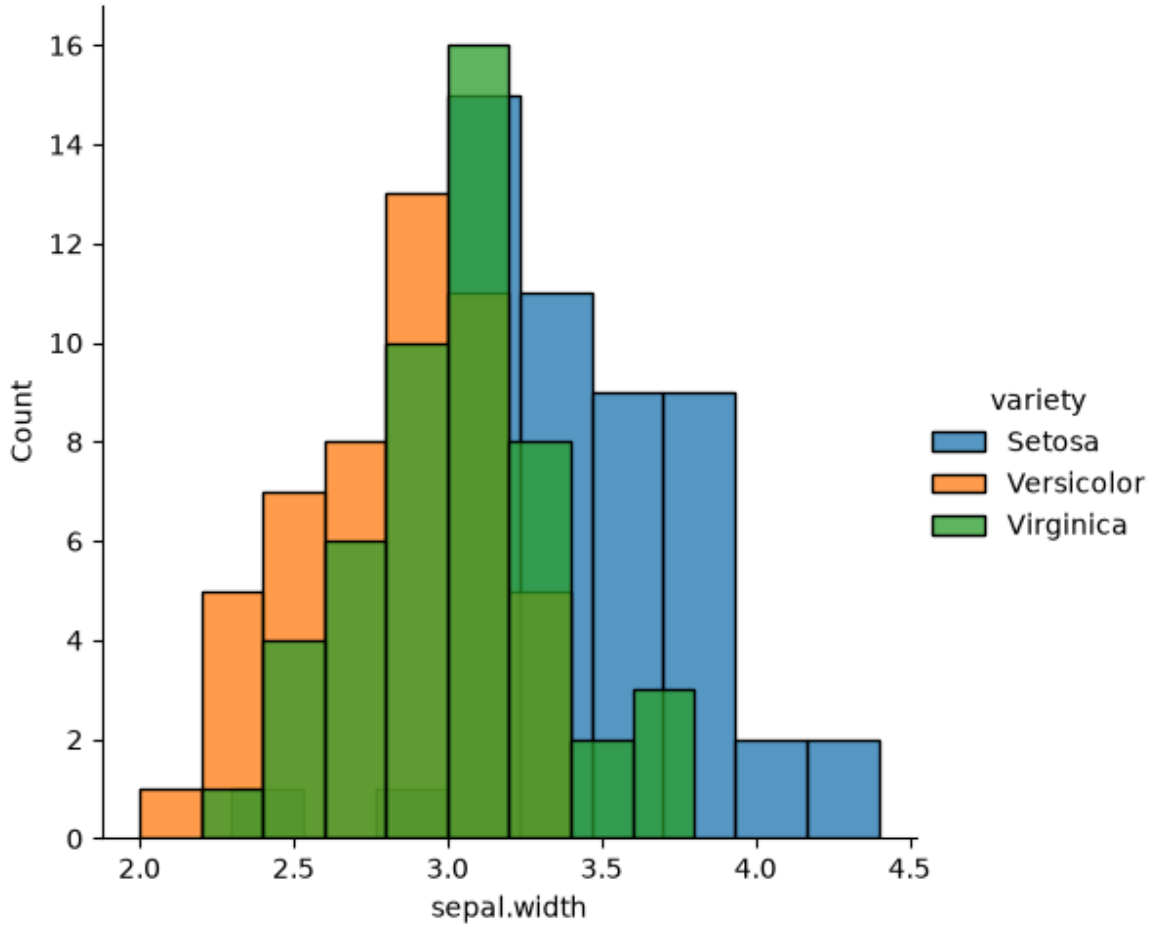
In [19]:
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'petal.width').add_l
plt.show();
```



In [20]:
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'sepal.length').add_
plt.show();
```



In [21]:
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'sepal.width').add_l
plt.show();
```


16
14
12
10
8
6
4
2
0
Count
2.0
2.5
3.0
3.5
4.0
4.5
sepal.width
variety
Setosa
Versicolor
Virginica

In [ ]: